Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “system” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

Two carts approach: one has momentum 5 kg·m/s east and the other 3 kg·m/s west.
Explain why adding their magnitudes gives the wrong total; find the signed total.

Word help: momentum = mass times velocity, including direction

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

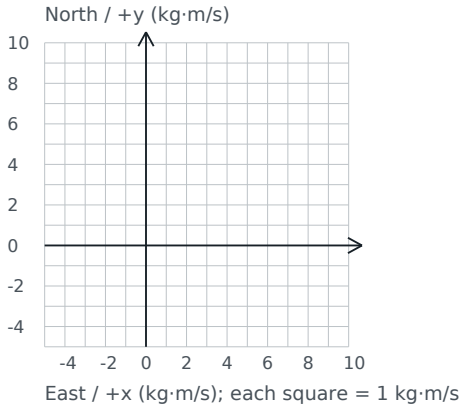
02 Worked example: explain the model

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$.
Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (3, 0) and (0, 4) kg·m/s. Their total is (3, 4), magnitude 5 kg·m/s. Add components before finding the magnitude.

Your turn: Draw a tip-to -tail momentum diagram and label each axis with units.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

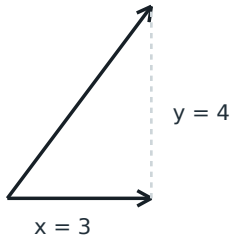
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two incoming arrows point east and north. **Sketch** the direction of their vector sum. Can their ordinary lengths be added for the diagonal magnitude?

Word help: magnitude = size of a quantity without its direction

Sentence starter: The … in my model represents … . Its label shows … .



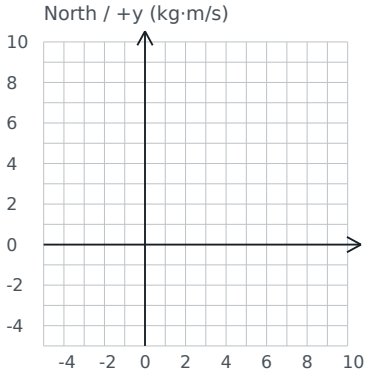
The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: 1 2 3

04 Apply the physics

Incoming total momentum is 6 kg·m/s east and 8 kg·m/s north. Two objects stick with total mass 2 kg. Divide each component by 2 to find velocity components.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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Continue down each column.

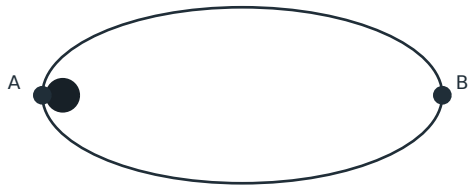
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

An ideal asteroid orbit has nearest distance 4 AU and farthest distance 8 AU. Where is it fastest?
Explain with the equal-area law.
Label the orbital features used in your explanation.

Word help: orbit = a path around another body under gravity

Sentence starter: My design shows … . It meets the requirement by … .

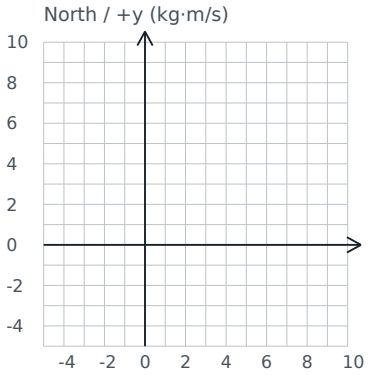


Marzano check - circle: ① ② ③

07 Exit check

Total momentum is 4 kg·m/s west and 2 kg·m/s south. Combined mass 2 kg.
Give final velocity components.

Sentence starter: I notice … . This supports … because … .

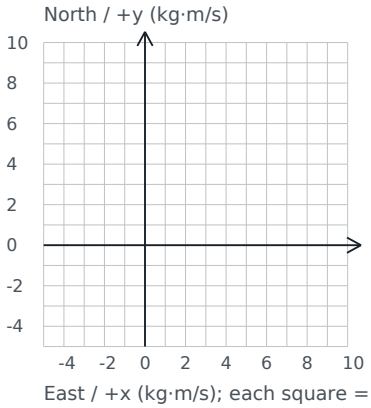


Marzano check - circle: ① ② ③

08 STRETCH: test the model

A joined pair of total mass 2 kg has total momentum 6 kg·m/s east and 8 kg·m/s north.
Find its velocity components using each momentum component divided by 2.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③